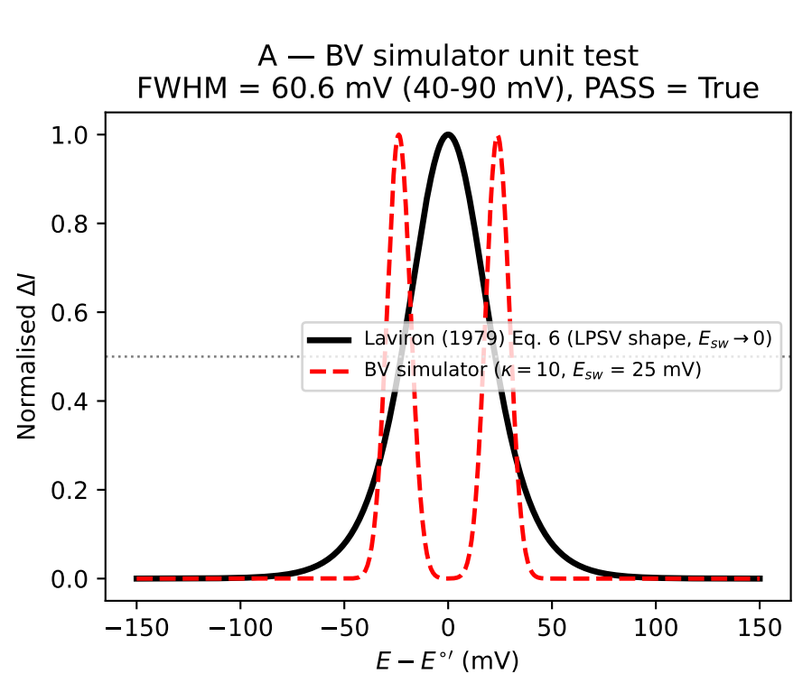
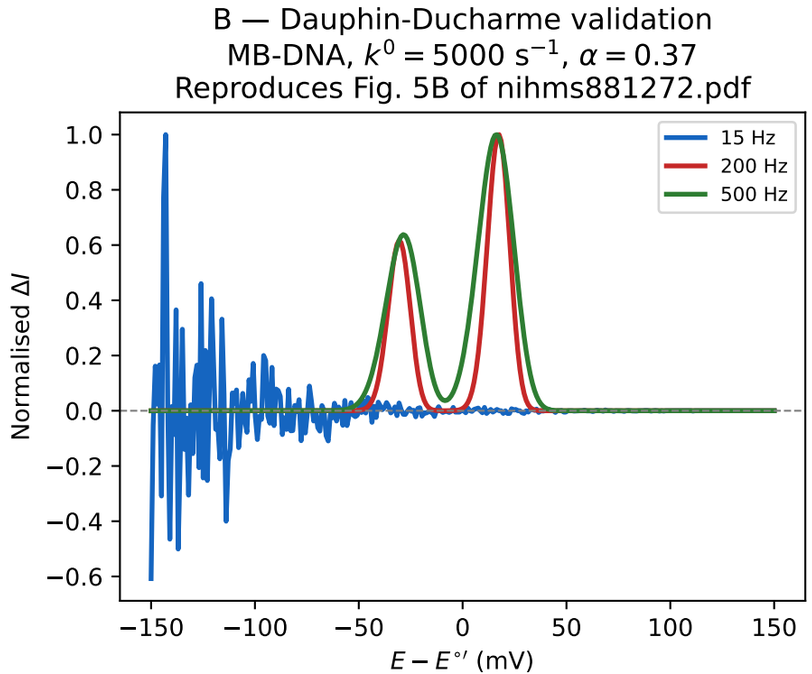
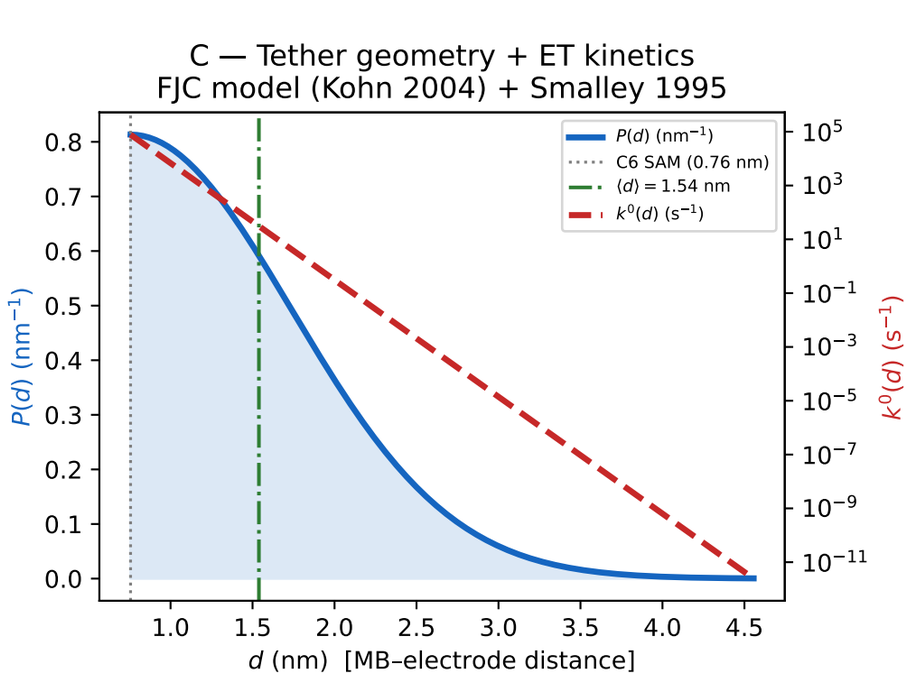

A — BV simulator unit test
FWHM = 60.6 mV (40-90 mV), PASS = True
Laviron (1979) Eq. 6 (LPSV shape, $E_{sw} \to 0$)
BV simulator ($\kappa = 10$, $E_{sw} = 25$ mV)
Normalised ΔI
$E - E^{\circ\prime}$ (mV)


B — Dauphin-Ducharme validation
MB-DNA, $k^0 = 5000$ s^{-1}, $\alpha = 0.37$
Reproduces Fig. 5B of nihms881272.pdf
15 Hz
200 Hz
500 Hz
Normalised ΔI
$E - E^{\circ\prime}$ (mV)


C — Tether geometry + ET kinetics
FJC model (Kohn 2004) + Smalley 1995
$P(d)$ (nm^{-1})
C6 SAM (0.76 nm)
$\langle d \rangle = 1.54$ nm
$k^0(d)$ (s^{-1})
d (nm) [MB–electrode distance]